



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

AGRICULTURAL SCIENCES P1

MAY/JUNE 2026

MARKING GUIDELINES

MARKS: 150

These marking guidelines consist of 11 pages.

SECTION A**QUESTION 1**

1.1	1.1.1	B ✓✓	(10 x 2)	(20)
	1.1.2	B ✓✓		
	1.1.3	D ✓✓		
	1.1.4	C ✓✓		
	1.1.5	D ✓✓		
	1.1.6	A ✓✓		
	1.1.7	C ✓✓		
	1.1.8	D ✓✓		
	1.1.9	B ✓✓		
	1.1.10	A ✓✓		
1.2	1.2.1	A only ✓✓	(5 x 2)	(10)
	1.2.2	None ✓✓		
	1.2.3	B only ✓✓		
	1.2.4	Both A and B ✓✓		
	1.2.5	A only ✓✓		
1.3	1.3.1	Chyme ✓✓	(5 x 2)	(10)
	1.3.2	Metabolic diseases ✓✓		
	1.3.3	Preparatory ✓✓		
	1.3.4	Buffer/pH regulator ✓✓		
	1.3.5	Congenital ✓✓		
1.4	1.4.1	Gross energy/GE ✓	(5 x 1)	(5)
	1.4.2	Crutching ✓		
	1.4.3	Placenta ✓		
	1.4.4	Seminiferous tubules ✓		
	1.4.5	Freemartin ✓		

TOTAL SECTION A: 45

SECTION B**QUESTION 2: ANIMAL NUTRITION****2.1 The alimentary canals of two farm animals**

2.1.1 Farm animal represented in DIAGRAM B
Fowl/chicken/poultry ✓ (1)

2.1.2 Justification of the answer

- Presence of a crop ✓
- Presence of proventriculus ✓
- Presence of ventriculus/gizzard ✓
- Presence of two caeca ✓ (Any 1) (1)

2.1.3 Description of protein digestion from part C/abomasum to part D/small intestine

- Protein is converted by enzyme pepsin into peptides in part C ✓ (1)
- The enzyme erepsin changes peptides to amino acids in part D ✓ (1)

2.1.4 ONE way in which micro-organisms in part B benefit animals

- Breakdown roughages/cellulose to produce volatile fatty acids/fermentation ✓
- Supply of water-soluble vitamins and vitamin K ✓
- They benefit from amino acids synthesized through hydrolysis of protein and non-protein nitrogenous compounds ✓ (Any 1) (1)

2.1.5 Comparing the oesophagus and the crop in terms of functions

Part A/oesophagus - Transports food from the mouth to the stomach and vice versa ✓ (1)

Part E/crop - Stores/moistens/softens food ✓ (1)

2.2 Feed component

2.2.1 Identification of the component of feed
Vitamins ✓ (1)

2.2.2 The category of vitamins
Water/fat soluble vitamins ✓ (1)

2.2.3 Indication of the letter of the component

(a) Preventing nervous symptoms in animals - C/D ✓ (1)

(b) Its deficiency causes polyneuritis - C ✓ (1)

2.2.4 Name of feed component labelled F
Vitamin A/retinol ✓ (1)

2.3 Digestibility of feeds

2.3.1 Explanation of what would have influenced the digestibility of maize stalk

The high fibre content in maize stalk was poorly digested (only 5kg) was absorbed ✓ 8kg was excreted by the young growing animal ✓ due to the underdevelopment of the rumen and insufficient rumen micro-organisms for fibre digestion ✓

OR

The animal was a non-ruminant ✓ and was less able to digest feeds high in crude fibre content ✓ due to a lack of rumen micro-organisms ✓

(3)

2.3.2 TWO methods that could be used to improve the digestibility

- Grinding ✓
- Pelleting ✓
- Spraying with water/NPN/molasses mixture ✓
- Supplementation with molasses/NPN ✓

(Any 2)

(2)

2.4 Types of feeds

2.4.1 Classification of feeds

(a) **PICTURE A** - Roughage ✓

(1)

(b) **PICTURE F** - Roughage ✓

(1)

2.4.2 Identification of feeds

(a) **A succulent roughage** - F (Silage)/A (Green Lucerne) ✓

(1)

(b) **A protein-rich concentrate of plant origin**

E (Soya bean meal) ✓

(1)

2.4.3 Explanation for feeding roughage to 8 week old calves

Roughages stimulate the development of the fore-stomach ✓ which promotes proper digestion of feed high in crude fibre and overall well-being of rumen ✓

(2)

2.4.4 Calculation of the nutritive ratio of the feed

$$\text{NR} = 1: \frac{\% \text{TDN} - \% \text{DP}}{\% \text{DP}} \quad \checkmark$$

$$\text{NR} = 1: \frac{85\% - 34\%}{34\%} \quad \checkmark$$

$$\text{NR} = 1:1,5/1:2 \quad \checkmark$$

OR

$$\text{NR} = 1: \frac{\% \text{DNNS}}{\% \text{DP}} \quad \checkmark$$

$$\text{NR} = 1: \frac{51\%}{34\%} \quad \checkmark$$

$$\text{NR} = 1:1,5/1:2 \quad \checkmark$$

(3)

2.5 Pearson square method**2.5.1 Name of the method**

Pearson square ✓

(1)

2.5.2 Identification of the ratioMaize meal : Peanut oilcake meal
27 : 8 ✓

(1)

2.5.3 Deduction of the amount of digestible protein for the mixture

17% ✓

(1)

2.6 Formulation of ration**2.6.1 Description of fodder flow programme**

A management strategy by farmers to ensure that livestock have sufficient feed ✓ throughout the year ✓

(2)

2.6.2 Type of the ration

Maintenance ration ✓

(1)

2.6.3 Calculation of the total monthly DM needed

- Per day: 25 cows x 10 kg DM/cow = 250 kg DM ✓
- 250 kg/day x 30 days ✓
- = 7 500 kg DM/month ✓

(3)

[35]**QUESTION 3: ANIMAL PRODUCTION, PROTECTION AND CONTROL****3.1 Production systems and farming systems****Identification of a farm**

(a) FARM D ✓

(1)

(b) FARM B ✓

(1)

(c) FARM A ✓

(1)

(d) FARM C ✓

(1)

3.2 Animal production**3.2.1 Identification of the factors to increase animal production**

(a) PICTURE A - Breeding/reproduction ✓

(1)

(b) PICTURE B - Nutrition/feeding ✓

(1)

3.2.2 Description of how nutrition can increase broiler production

Proper nutrition gives farm animals vigour to grow/develop/strong immunity to fight off infections ✓ leading to more profitable and sustainable agriculture ✓

(2)

3.3 Effects of incorrect handling of farm animals

- 3.3.1 A ✓ (1)
- 3.3.2 C ✓ (1)
- 3.3.3 B ✓ (1)

3.4 TWO basic guidelines to be considered before loading animals

- Familiarise animals with loading area ✓
- Group animals before loading them so that they establish social groups ✓
- Ensure that the truck/loading zone is suitable for the type of animal ✓
- Keep animals calm in preparation for the trip ✓
- Do not feed animals two hours before they are loaded ✓
- Separate animals according to type, sex and age ✓
- Separate sick/pregnant animals ✓
- Floors should not be slippery ✓
- Disinfect the truck prior to loading animals ✓
- Proper documentation must be available ✓ (Any 2) (2)

3.5 Housing facility**3.5.1 Purpose of a broiler housing facility**

- To create a suitable environment for maximum growth and production ✓
- Protection against predators ✓ (Any 1) (1)

3.5.2 ONE basic requirement of the housing facility

- House is oriented lengthwise in a north-south direction and short sides facing east -west ✓
- Proper lighting ✓
- Side walls built in such a way to allow for air movement ✓
- Corrugated iron increases temperature during winter months ✓
- House have the foldable curtains to regulate temperature ✓
- Roof has an A-frame to improve air circulation ✓ (Any 1) (1)

3.5.3 TWO equipment to reduce heat stress in summer

- Air conditioner/evaporative cooling pads ✓
- Foggers/mist sprays ✓
- Fans ✓
- Foldable curtains/ventilators ✓ (Any 2) (2)

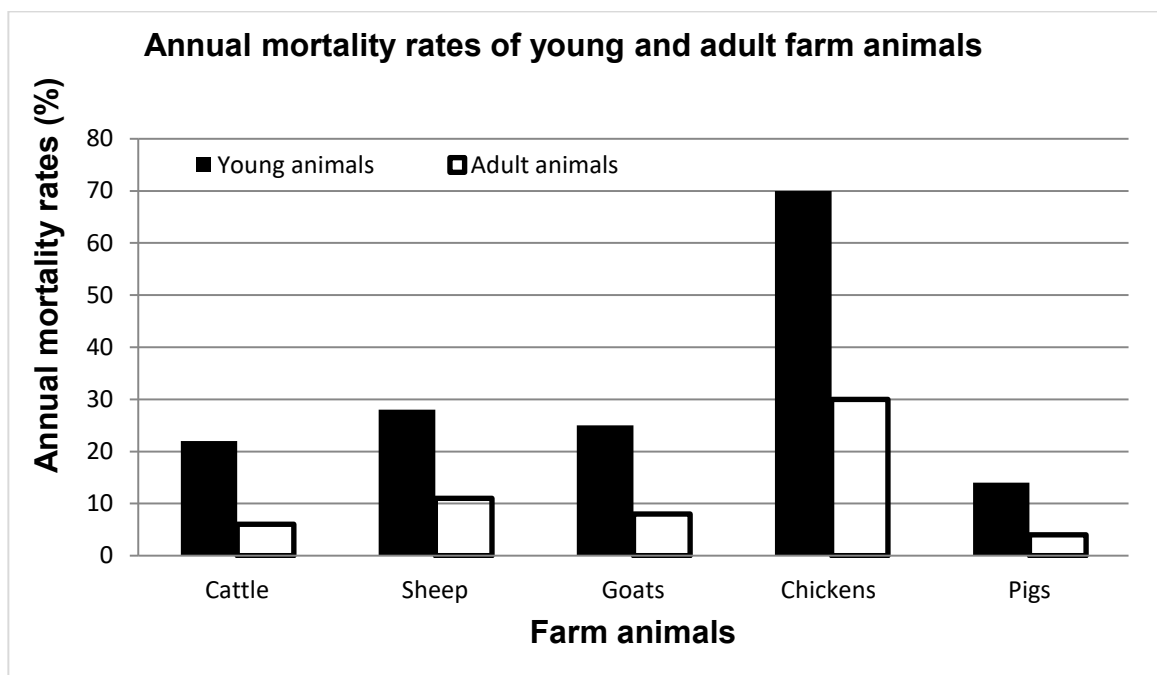
3.6 Health and mortality rates in farm animals

3.6.1 ONE general sign of ill-health in pigs

- Loss of appetite ✓
- Isolation from the group ✓
- Listlessness ✓
- Swollen navel, udder or joints ✓
- Rapid, irregular breathing ✓
- Shivering ✓
- Coughing ✓
- Discoloured skin/blisters ✓
- Loss/lack of condition/weight ✓
- Lameness/lethargy ✓
- Diarrhoea/constipation ✓
- Vomiting ✓

(Any 1) (1)

3.6.2 Combined bar graph



Criteria/rubric/marketing guidelines

- Correct heading (with both variables) ✓
- X-axis: Correctly calibrated with label (Farm animal) ✓
- Y-axis: Correctly calibrated with label (Annual mortality rates) ✓
- Correct units (%) ✓
- Combined bar graph ✓
- Accuracy (80% + correctly plotted) ✓

(6)

3.7 Transmission and prevention of animal diseases**3.7.1 Mode of transmission**

- (a) **Anthrax** - Inhalation/direct contact/ingestion ✓ (1)
- (b) **Rift Valley Fever** - Vectors ✓ (1)

3.7.2 TWO ways to prevent further spread of diseases once detected

- Vaccination ✓
- Isolation/quarantine ✓
- Controlling vectors ✓
- Proper disposal of carcasses and bedding material ✓
- Notifying the relevant authorities ✓
- Disinfection/sanitation ✓ (Any 2) (2)

3.8 Parasites**3.8.1 Collective term of the organisms**

Parasites ✓ (1)

3.8.2 Identification of the parasites

- (a) **Needs a snail to complete its life cycle** - PICTURE A ✓ (1)
- (b) **A vector** - PICTURE C ✓ (1)

3.8.3 Differentiation of organisms

- **PICTURE A:** Found inside the body of the host/internally ✓ (1)
- **PICTURE C:** Found outside the body of the host/externally ✓ (1)

3.9 Salt poisoning**3.9.1 ONE measure to treat salt poisoning**

- Internal provision of fresh water in small quantities through stomach tube ☐
- Treat animals with a hypertonic dextrose or isotonic saline solution ☐
- Immediate removal of the source of salt poisoning ☐ (Any 1) (1)

3.9.2 ONE role of the state in animal protection

- Public awareness campaigns ✓
 - Provision of vaccines ✓
 - Provision of veterinary services ✓
 - Provision of quarantine services ✓
 - Conducts research on animal diseases/pests ✓
 - Provides advice, monitoring and control ✓
 - Formulates, approves and monitors the implementation of legislation ✓
 - Regulates imports/exports ✓ (Any 1) (1)
- [35]**

QUESTION 4: ANIMAL REPRODUCTION**4.1 Female reproductive system**

4.1.1 The process represented by A
Ovulation ✓ (1)

4.1.2 Justify the answer
The egg cell is released from the Graafian follicle/ovary ✓ (1)

4.1.3 Description of the process in B/fertilisation

- Process when a haploid male gamete/sperm cell and a haploid female gamete/ovum fuse ✓ to form a diploid zygote ✓
- Fusion of male and female gametes ✓ to form diploid zygote ✓ (Any 1) (2)

4.1.4 Type of cell division by which the zygote divides multiple times
Mitosis ✓ (1)

4.1.5 ONE function of the uterus

- Provides a suitable environment for development of the foetus ✓
- Helps with the transportation of sperm cells ✓
- Helps with the expulsion of the calf ✓
- Provides uterine milk as nutrition to the developing embryo ✓
- Contains nodules where foetal membranes are attached to ✓
- Sperm capacitation ✓ (Any 1) (1)

4.2 Re-arrangement of the mating stages in sequential order

B ✓ (1)

C ✓ (1)

A ✓ (1)

4.3 Hormone concentration during the reproductive cycle in cows

4.3.1 Identification of the progesterone concentration on day 12
30 mg/ml ✓ (1)

4.3.2 Hormones responsible

(a) Oestrogen/progesterone ✓ (1)

(b) Luteinising hormone/LH ✓ (1)

4.3.3 Description of the change in the progesterone concentration
Progesterone concentration increase/will become higher ✓ (1)

- 4.3.4 **Identification of the oestrus cycle stage**
Di-oestrus ✓ (1)
- 4.3.5 **The impact of progesterone during di-oestrus**
- Progesterone suppresses the secretion of follicle stimulating hormone (FSH) ✓ to prevent the development of follicles/onset of oestrus ✓
 - Maintains the endometrium ✓ in preparation for implantation ✓ (Any 1) (2)
- 4.4 **Morphological abnormalities in sperm cells**
- 4.4.1 **The morphological abnormalities in sperm cells**
B/C ✓ (1)
- 4.4.2 **Identification of the number of the part responsible for**
- (a) **Mobility** - 3 ✓ (1)
- (b) **Carrier of the genetic information** - 1 □ (1)
- 4.4.3 **Structure releasing the enzyme responsible for sperm penetration**
Acrosome □ (1)
- 4.4.4 **The process whereby sperm cells are formed**
Spermatogenesis □ (1)
- 4.5 **Reproductive techniques in farm animals**
- 4.5.1 **Identification of the reproductive technique**
Embryo transfer/transplant/ET ✓ (1)
- 4.5.2 **Importance of the recipient cow**
Implantation of the embryos/acts as a surrogate ✓ (1)
- 4.5.3 **Definition of cloning**
Process of generating genetically identical copy of cell/organism ✓✓ (2)
- 4.5.4 **Identification of the sheep**
Nucleus donor sheep ✓ (1)
- 4.5.5 **Justification of the answer**
Genetical material is from somatic cells of the nucleus donor sheep ✓ (1)
- 4.6 **Pregnancy**
- 4.6.1 **Deduction the cow's stage of pregnancy**
Foetal stage ✓ (1)
- 4.6.2 **Description of embryonic stage of pregnancy**
It is the stage after the ovum stage ✓ during which organs, tissues and systems develop ✓ (2)

4.6.3 TWO calf-related possible causes of birth difficulty

- Abnormal large size foetus/hydropsy ✓
- Multiple births/twins ✓
- Incorrect/posterior presentation of the foetus/deviation of the head/flexion of the elbow ✓
- Hydrocephalus ✓

(Any 2) (2)

4.7 Lactation**4.7.1 The period of lactation during which milk production declines**

After week 8 to week 9 ✓

(1)

4.7.2 Explanation of the relationshipMilk production declines while butterfat content increases/
vice versa ✓✓

(2)

[35]**TOTAL SECTION B: 105**
GRAND TOTAL: 150